

LABPULSE LABORATORY MANAGEMENT SYSTEM

BECS 31242 – Visual Programming

Group No -13

Group Members :

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Introduction

The LabPulse Laboratory Resource Management System is an integrated software platform designed to automate, monitor, and streamline the lifecycle of inventory assets and checkout workflows within university electronics and computer science environments. Manual tracking of high-demand test equipment, prototyping modules, and discrete components often results in inventory discrepancies, untracked damage liabilities, and administrative backlogs.

LabPulse bridges this gap by establishing a reliable, relational database architecture using a C# Windows Forms desktop frontend paired with a centralized MySQL server. The system enforces clear workflow separations between staff administrators and students, ensuring precise item serialization, immediate stock level monitoring, automated reservation planning, and an accountable damage/fine resolution framework. By bringing real-time precision to laboratory resource tracking, LabPulse maximizes equipment availability and protects institutional assets.

Main System Functionalities

- **User Authentication & Role-Based Access Control**

New users can provision account profiles by providing registration criteria, including name, unique email address validation checks, secure passwords, and dedicated structural roles

Dynamic ID Generation which the backend system auto-generates unique 6-digit UserID sequence tracking markers upon registration to avoid account collision



Registration

CREATE ACCOUNT

Your Name
Amal Jayanath

Your Email
amal@gmail.com

Password
12345

Re enter passwaord
12345

Role (Staff/Student)
staff

SIGN UP

Have already an account? [Login here](#)

Registration Success

i Account successfully created!

OK

- **Inventory Registry & Stock Maintenance**

Authorized personnel can catalog new hardware pieces, test accessories, or workspace assets by providing an Equipment Name, explicit ID fields, and an Initial Stock Count.

Provides administrative maintenance tools allowing staff to dynamically execute data modifications, synchronize backend adjustments (LOAD, UPDATE), or purge deprecated items from the pool.

The screenshot shows a web browser window titled "ManageEquipment". The main heading is "Equipment Registry and Stock Controller". There are two links: "Back" on the left and "Home" on the right. The interface is divided into two main sections. The first section, "Register new Equipment Here :", contains three input fields: "Equipment Name", "Equipment ID", and "Initial Stock Count". Below these fields is a red button labeled "REGISTER ASSET". The second section, "Update, Add , Delete Equipment Here :", contains a large gray rectangular area. At the bottom of the interface are three red buttons: "LOAD", "CLEAR", and "UPDATE ASSET".

ManageEquipment

Equipment Registry and Stock Controller

[Back](#) [Home](#)

Register new Equipment Here :

Equipment Name

Equipment ID

Initial Stock Count

REGISTER ASSET

Update, Add , Delete Equipment Here :

LOAD **CLEAR** **UPDATE ASSET**

- **Resource Allocation Processing (Check-Out Workflow)**

acts as a live control center tracking incoming item requests, displaying a live numerical gauge of outstanding allocations awaiting processing. acts as a live control center tracking incoming item requests, displaying a live numerical gauge of outstanding allocations awaiting processing.

Renders detailed structural views of all logged transaction timelines, recording

Selecting an entry in the grid auto-populates validation textboxes. Clicking the validation button approves the check-out process, seamlessly modifying remaining asset counts in the database

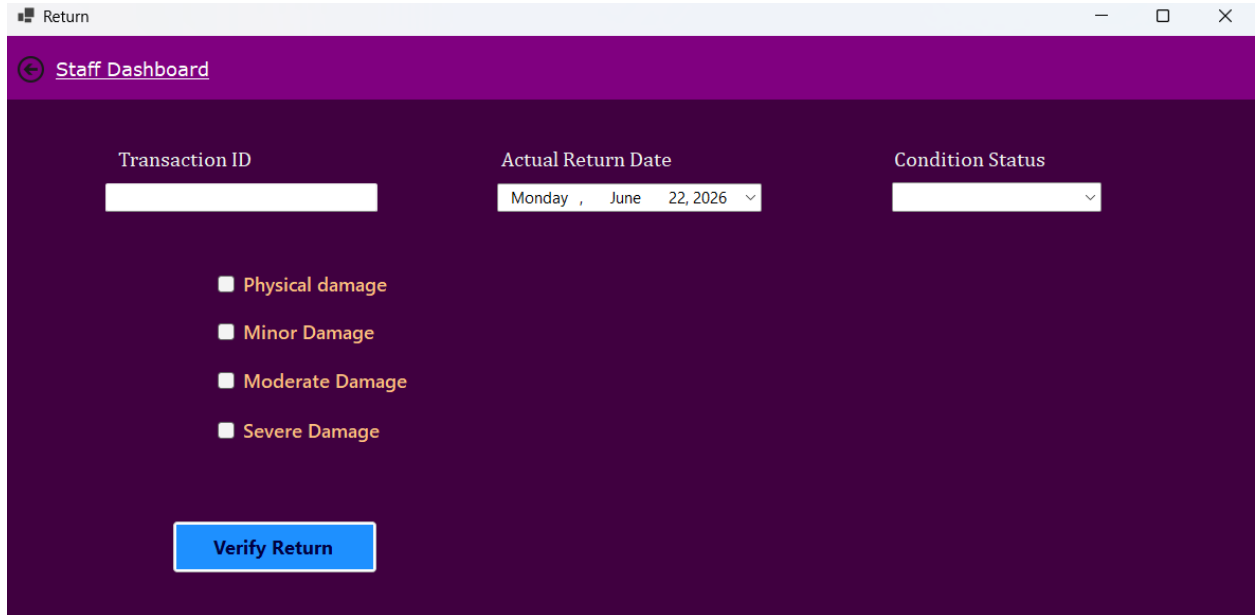
The screenshot shows a web application window titled 'Issue'. The main header is 'Staff Dashboard'. The title of the section is 'Process Resource Allocation Check-Out'. Below the title, there is a status indicator: 'No of Pending Requests 3'. The form contains five input fields: 'Student ID' (value: 1), 'Student Name' (value: Student Profile), 'Equipment Name' (value: Resource Asset), 'Equipment ID' (value: 4), and 'Allocation Date (Start)' (value: 2026-06-20). Below the form are two buttons: 'Approve Issue Request' and 'Refresh'. At the bottom, there is a table with 8 columns: TransactionID, UserID, EquipmentID, Quantity, IssueDate, ActualReturnDate, and Status. The table contains 6 rows of data, with the third row (TransactionID: 12009) highlighted in blue.

	TransactionID	UserID	EquipmentID	Quantity	IssueDate	ActualReturnDate	Status
	67	535377	7	5	6/21/2026	7/6/2026	Returned
	456	1	3	1	6/3/2026	6/26/2026	Returned
▶	12009	1	4	1	6/20/2026	7/4/2026	Pending
	61832	1	6	5	6/24/2026	7/8/2026	Pending
	86479	1	1	1	6/21/2026	7/5/2026	Pending

- **Structural Inspection & Asset Return Desk**

When equipment is physically presented for return, administrators evaluate items against multi-tiered classification rules (Perfect, Physical, Minor, Moderate, or Severe Damage states)

If an asset is returned in satisfactory condition (e.g., "Perfect" or "Minor Damage"), the quantities are automatically calculated and returned to the active inventory pool.

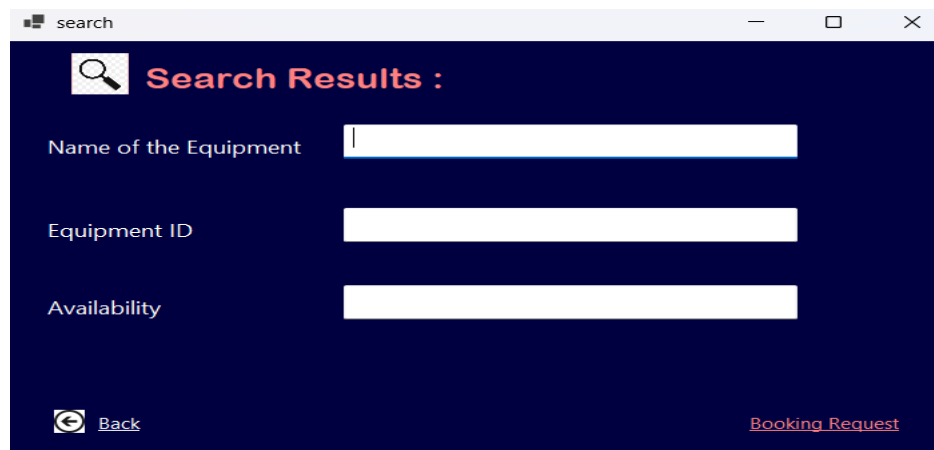


The screenshot shows a web application window titled "Return". The interface has a purple header bar with a back arrow and the text "Staff Dashboard". Below the header, there are three input fields: "Transaction ID" (a text box), "Actual Return Date" (a date picker showing "Monday, June 22, 2026"), and "Condition Status" (a dropdown menu). Under these fields, there are four radio button options: "Physical damage", "Minor Damage", "Moderate Damage", and "Severe Damage". At the bottom of the form is a blue button labeled "Verify Return".

- **Equipment Discovery & Smart Search**

Students can query the laboratory inventory using specific string attributes to quickly identify item parameters

The system returns structural details showing specific component names, active identification numbers, and real-time inventory status markers



The screenshot shows a web application window titled "search". The interface has a dark blue header bar with a magnifying glass icon and the text "Search Results :". Below the header, there are three input fields: "Name of the Equipment" (a text box), "Equipment ID" (a text box), and "Availability" (a text box). At the bottom of the form, there is a "Back" button with a circular arrow icon and a "Booking Request" link.

- **Equipment Allocation & Booking**

Allows students to submit instant resource checkouts where the allocation launch date is fixed automatically to the current calendar date.

Allows students to plan future projects by specifying a custom date using an embedded calendar picker tool. Both operations append secure rows to the tracking tables under an explicit Pending status until approved by staff

The screenshot shows a web browser window titled "reservation". The page has a purple header with a "Student Dashboard" link. Below the header, there are two main sections for booking equipment.

BOOK NOW

This section contains three input fields: "Equipment ID *", "Email Address", and "Qty". To the right of these fields is a "Book Now" button.

RESERVE A BOOKING

This section contains four input fields: "Equipment ID *", "Email Address", "Qty", and "Reserve Date (Start)". The "Reserve Date (Start)" field is a date picker showing "Monday , June 22, 2026". To the right of these fields is a "Proceed to Reserve" button.

At the bottom of the page, there is a link: [* Look at the Equipment ID Here](#)

Data tablesp

User Table

← ↗ → ▼										UserID	Name	Email	Password	Role	PhoneNo	Address
☐				1	Ravindu Prabashana	prabashana931@gmail.com	abc123	admin	0742502695	186/a/1, Udakanampella, Pugoda						
☐				108520	Nimesh Dilshan	nimesh23@gmail.com	pqr123	student	0719230431	deniyaya, Matara						
☐				211162	a	a@a.a	a	student								
☐				233127	a	iuehfwfh@dj.fs	a	staff								
☐				486201	Sithum Udara	sthudara@gmail.com	123456	admin								
☐				535377	Umayanga	umayanga@gmail.com	def123	admin	0775698846	geradi ella, Kurunegala						
☐				547810	a	a@a.a	a	admin								
☐				887437	Amal Jayanath	amal@gmail.com	12345	staff								

Equipment table

			EquipmentID	Name	Remaining Quantity	Status
☐				1	ESP32 Dev kit	5 In Stock
☐				2	Florence Flask	0 Out of Stock
☐				3	Arduino Uno Board	3 In Stock
☐				4	Arduino Nano Board	13 In Stock
☐				5	Digital Oscilloscope	2 In Stock
☐				6	RED LED	65 In Stock
☐				7	BLUE LED	5 In Stock
☐				8	Ammeter	3 In Stock
☐				9	Volt Meter	3 In Stock

Transaction table

				TransactionID	UserID	EquipmentID	Quantity	IssueDate	ActualReturnDate	Status
☐				67	535377	7	5	2026-06-21	2026-07-06	Returned
☐				456	1	3	1	2026-06-03	2026-06-26	Returned
☐				12009	1	4	1	2026-06-20	2026-07-04	Pending
☐				61832	1	6	5	2026-06-24	2026-07-08	Pending
☐				86479	1	1	1	2026-06-21	2026-07-05	Pending
	☐ Check all	With selected:					Export			

Contribution of each group member

Member	Responsibility	Forms
EC/2022/041	Authentication & Dashboard	Login Form, Dashboard
EC/2022/065	User Management	Registration, Profile Manager
EC/2022/079	Inventory Management	Category Manager, Equipment Registry
EC/2022/031	Transactions & Reports	Issue, Return, Reports